Supplementary Material for "Comparison of Wind-related Storage Investment Options in a Market-based Environment: Direct Ownership, Cooperative, and Competitive Modes"

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The strong duality equality of the problem (a.6)-(a.27) can be written as follows:

$$\sum_{s\in S}\pi_s\sum_{t\in T}\Big(\sum_{k\in K}\big(C_k^{\mathrm{g}}p_{s,k,t}^{\mathrm{g}}+C_k^{\mathrm{g}\uparrow}r_{s,k,t}^{\mathrm{g}\uparrow}+C_k^{\mathrm{g}\downarrow}r_{s,k,t}^{\mathrm{g}\downarrow}\big)+\sum_{u\in U}o_{s,u,t}^{\mathrm{w}}p_{s,u,t}^{\mathrm{w}}+\sum_{i\in I}\big(o_{s,i,t}^{\mathrm{dis}}p_{s,i,t}^{\mathrm{dis}}-o_{s,i,t}^{\mathrm{ch}}p_{s,i,t}^{\mathrm{ch}}+o_{s,i,t}^{\mathrm{ch}\uparrow}r_{s,i,t}^{\mathrm{ch}\uparrow}+o_{s,i,t}^{\mathrm{dis}\uparrow}r_{s,i,t}^{\mathrm{dis}\uparrow}+o_{s,i,t}^{\mathrm{ch}\downarrow}r_{s,i,t}^{\mathrm{ch}\downarrow}+o_{s,i,t}^{\mathrm{dis}\downarrow}r_{s,i,t}^{\mathrm{dis}\downarrow}\big)\Big)=\sum_{s\in S}\pi_s\sum_{t\in T}\Big(-\sum_{u\in U}\overline{\alpha}_{s,u,t}^{\mathrm{w}}P_{s,u,t}^{\mathrm{wf}}-\sum_{i\in I}\Big(\big(\overline{\alpha}_{s,i,t}^{\mathrm{ch}}+\overline{\alpha}_{s,i,t}^{\mathrm{dis}}+\overline{\alpha}_{s,i,t}^{\mathrm{dis}\uparrow}+\overline{\alpha}_{s,i,t}^{\mathrm{ch}\downarrow}\big)p_i^{\max}+\overline{\alpha}_{s,i,t}^{\mathrm{e}}e_i^{\max}\Big)\Big)+\kappa \quad \text{(e.1)}$$

where $\kappa=\sum_{s\in S}\pi_s\sum_{t\in T}\Big(-\sum_{k\in K}\overline{\alpha}_{s,k,t}^{\mathrm{g}}P_k^{\mathrm{g,max}}+\sum_{i\in I}\big(\lambda_{s,i,t}\mathrm{D}_{i,t}-\overline{\alpha}_{s,i,t}^{\mathrm{ang}}\theta_{\max}+\underline{\alpha}_{s,i,t}^{\mathrm{ang}}\theta_{\min}\big)+\lambda_{s,t}^{\uparrow}\mathrm{R}_{s,t}^{\uparrow}+\lambda_{s,t}^{\downarrow}\mathrm{R}_{s,t}^{\downarrow}-\sum_{(i,j)\in L}\Big(\underline{\alpha}_{s,ij,t}^{\mathrm{l}}F_{ij}^{\max}+\overline{\alpha}_{s,ij,t}^{\mathrm{l}}F_{ij}^{\max}\Big)\Big)$

From (d.31)–(d.37), $\lambda_{s,i,t}$, $\lambda_{s,t}^{\uparrow}$, and $\lambda_{s,t}^{\downarrow}$ can be expressed as:

$$\lambda_{s,i(u\in\Omega_i),t}=o_{s,u,t}^{\mathrm{w}}+\overline{\alpha}_{s,u,t}^{\mathrm{w}}-\underline{\alpha}_{s,u,t}^{\mathrm{w}} \quad \text{(e.2)}$$
$$\lambda_{s,i,t}=o_{s,i,t}^{\mathrm{ch}}-\overline{\alpha}_{s,i,t}^{\mathrm{ch}}+\underline{\alpha}_{s,i,t}^{\mathrm{ch}}+\overline{\alpha}_{s,i,t}^{\mathrm{ch}\uparrow}-\overline{\alpha}_{s,i,t}^{\mathrm{ch}\downarrow}+\eta_c\alpha_{s,i,t}^{\mathrm{e}} \quad \text{(e.3)}$$
$$\lambda_{s,i,t}=o_{s,i,t}^{\mathrm{dis}}+\overline{\alpha}_{s,i,t}^{\mathrm{dis}}-\underline{\alpha}_{s,i,t}^{\mathrm{dis}}+\overline{\alpha}_{s,i,t}^{\mathrm{dis}\uparrow}-\overline{\alpha}_{s,i,t}^{\mathrm{dis}\downarrow}+\alpha_{s,i,t}^{\mathrm{e}}/\eta_d \quad \text{(e.4)}$$
$$\lambda_{s,t}^{\uparrow}=o_{s,i,t}^{\mathrm{ch}\uparrow}+\overline{\alpha}_{s,i,t}^{\mathrm{ch}\uparrow}-\underline{\alpha}_{s,i,t}^{\mathrm{ch}\uparrow}+\eta_c\underline{\alpha}_{s,i,t}^{\mathrm{e}} \quad \text{(e.5)}$$
$$\lambda_{s,t}^{\uparrow}=o_{s,i,t}^{\mathrm{dis}\uparrow}+\overline{\alpha}_{s,i,t}^{\mathrm{dis}\uparrow}-\underline{\alpha}_{s,i,t}^{\mathrm{dis}\uparrow}+\underline{\alpha}_{s,i,t}^{\mathrm{e}}/\eta_d \quad \text{(e.6)}$$
$$\lambda_{s,t}^{\downarrow}=o_{s,i,t}^{\mathrm{ch}\downarrow}+\overline{\alpha}_{s,i,t}^{\mathrm{ch}\downarrow}-\underline{\alpha}_{s,i,t}^{\mathrm{ch}\downarrow}+\eta_c\overline{\alpha}_{s,i,t}^{\mathrm{e}} \quad \text{(e.7)}$$
$$\lambda_{s,t}^{\downarrow}=o_{s,i,t}^{\mathrm{dis}\downarrow}+\overline{\alpha}_{s,i,t}^{\mathrm{dis}\downarrow}-\underline{\alpha}_{s,i,t}^{\mathrm{dis}\downarrow}+\overline{\alpha}_{s,i,t}^{\mathrm{e}}/\eta_d \quad \text{(e.8)}$$

From (d.5)–(d.18):

$$\big(\overline{\alpha}_{s,u,t}^{\mathrm{w}}-\underline{\alpha}_{s,u,t}^{\mathrm{w}}\big)p_{s,u,t}^{\mathrm{w}}=P_{s,u,t}^{\mathrm{wf}}\overline{\alpha}_{s,u,t}^{\mathrm{w}} \quad \text{(e.9)}$$
$$\Big(\overline{\alpha}_{s,i,t}^{\mathrm{dis}}-\underline{\alpha}_{s,i,t}^{\mathrm{dis}}\Big)p_{s,i,t}^{\mathrm{dis}}=\overline{\alpha}_{s,i,t}^{\mathrm{dis}}p_i^{\max} \quad \text{(e.10)}$$
$$\Big(\overline{\alpha}_{s,i,t}^{\mathrm{ch}}-\underline{\alpha}_{s,i,t}^{\mathrm{ch}}\Big)p_{s,i,t}^{\mathrm{ch}}=\overline{\alpha}_{s,i,t}^{\mathrm{ch}}p_i^{\max} \quad \text{(e.11)}$$
$$\Big(\overline{\alpha}_{s,i,t}^{\mathrm{ch}\uparrow}-\underline{\alpha}_{s,i,t}^{\mathrm{ch}\uparrow}\Big)r_{s,i,t}^{\mathrm{ch}\uparrow}=p_{s,i,t}^{\mathrm{ch}}\overline{\alpha}_{s,i,t}^{\mathrm{ch}\uparrow} \quad \text{(e.12)}$$
$$\Big(\overline{\alpha}_{s,i,t}^{\mathrm{dis}\uparrow}-\underline{\alpha}_{s,i,t}^{\mathrm{dis}\uparrow}\Big)r_{s,i,t}^{\mathrm{dis}\uparrow}=\big(p_i^{\max}-p_{s,i,t}^{\mathrm{dis}}\big)\overline{\alpha}_{s,i,t}^{\mathrm{dis}\uparrow} \quad \text{(e.13)}$$
$$\Big(\overline{\alpha}_{s,i,t}^{\mathrm{dis}\downarrow}-\underline{\alpha}_{s,i,t}^{\mathrm{dis}\downarrow}\Big)r_{s,i,t}^{\mathrm{dis}\downarrow}=p_{s,i,t}^{\mathrm{dis}}\overline{\alpha}_{s,i,t}^{\mathrm{dis}\downarrow} \quad \text{(e.14)}$$
$$\Big(\overline{\alpha}_{s,i,t}^{\mathrm{ch}\downarrow}-\underline{\alpha}_{s,i,t}^{\mathrm{ch}\downarrow}\Big)r_{s,i,t}^{\mathrm{ch}\downarrow}=\big(p_i^{\max}-p_{s,i,t}^{\mathrm{ch}}\big)\overline{\alpha}_{s,i,t}^{\mathrm{ch}\downarrow} \quad \text{(e.15)}$$

Using (e.2)–(e.15), the objective function (a.1) can be rewritten as:

$$\sum_{t\in T}\sum_{u\in U}\big(o_{s,u,t}^{\mathrm{w}}p_{s,u,t}^{\mathrm{w}}+P_{s,u,t}^{\mathrm{wf}}\overline{\alpha}_{s,u,t}^{\mathrm{w}}\big)+\sum_{i\in I}\Big(o_{s,i,t}^{\mathrm{dis}}p_{s,i,t}^{\mathrm{dis}}-o_{s,i,t}^{\mathrm{ch}}p_{s,i,t}^{\mathrm{ch}}+o_{s,i,t}^{\mathrm{ch}\uparrow}r_{s,i,t}^{\mathrm{ch}\uparrow}+o_{s,i,t}^{\mathrm{dis}\uparrow}r_{s,i,t}^{\mathrm{dis}\uparrow}+o_{s,i,t}^{\mathrm{ch}\downarrow}r_{s,i,t}^{\mathrm{ch}\downarrow}+o_{s,i,t}^{\mathrm{dis}\downarrow}r_{s,i,t}^{\mathrm{dis}\downarrow}+\Big(\overline{\alpha}_{s,i,t}^{\mathrm{dis}\uparrow}+\overline{\alpha}_{s,i,t}^{\mathrm{ch}\downarrow}+\overline{\alpha}_{s,i,t}^{\mathrm{ch}}+\overline{\alpha}_{s,i,t}^{\mathrm{dis}}\Big)p_i^{\max}+\big(r_{s,i,t}^{\mathrm{dis}\downarrow}/\eta_d+\eta_c r_{s,i,t}^{\mathrm{ch}\downarrow}\big)\overline{\alpha}_{s,i,t}^{\mathrm{e}}+\big(r_{s,i,t}^{\mathrm{dis}\uparrow}/\eta_d+\eta_c r_{s,i,t}^{\mathrm{ch}\uparrow}\big)\underline{\alpha}_{s,i,t}^{\mathrm{e}}+\big(p_{s,i,t}^{\mathrm{dis}}/\eta_d-\eta_c p_{s,i,t}^{\mathrm{ch}}\big)\alpha_{s,i,t}^{\mathrm{e}}\Big) \quad \text{(e.16)}$$

From (d.19), (d.20), (d.29), (d.30):

$$\sum_{t\in T}\sum_{i\in I}\Big(\big(r_{s,i,t}^{\mathrm{dis}\downarrow}/\eta_d+\eta_c r_{s,i,t}^{\mathrm{ch}\downarrow}\big)\overline{\alpha}_{s,i,t}^{\mathrm{e}}+\big(r_{s,i,t}^{\mathrm{dis}\uparrow}/\eta_d+\eta_c r_{s,i,t}^{\mathrm{ch}\uparrow}\big)\underline{\alpha}_{s,i,t}^{\mathrm{e}}+\big(p_{s,i,t}^{\mathrm{dis}}/\eta_d-\eta_c p_{s,i,t}^{\mathrm{ch}}\big)\alpha_{s,i,t}^{\mathrm{e}}\Big)=\sum_{t\in T}\sum_{i\in I}\big(e_i^{\max}\overline{\alpha}_{s,i,t}^{\mathrm{e}}\big) \quad \text{(e.17)}$$

Substituting (e.1), (e.17), in (e.16), the linearized form of the objective function (a.1) can be obtained:

$$\sum_{s\in S}\pi_s\sum_{t\in T}\sum_{k\in K}\Big(-C_k^{\mathrm{G}}p_{s,k,t}^{\mathrm{g}}-C_k^{\mathrm{g}\uparrow}r_{s,k,t}^{\mathrm{g}\uparrow}-C_k^{\mathrm{g}\downarrow}r_{s,k,t}^{\mathrm{g}\downarrow}\Big)+\kappa \quad \text{(e.18)}$$

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